Chaebin Jeong

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RESEARCH INTERESTS

Knowledge-graph-based data generation (knowledge graph construction, LLM training data synthesis), LLM evaluation benchmarks, diagnosing and improving time-series foundation models (TSFMs), LLM agent memory, and food recommendation systems.

EDUCATION

M.S. in Artificial Intelligence and Data Science Sep 2025 – Aug 2027 (expected)
Sejong University, Seoul, Korea GPA 4.38/4.50

- Food and Nutrition AI (FNAI) Lab, advised by Prof. Donghyeon Park

B.S. in Electronics and Information Engineering Mar 2020 – Aug 2025
Sejong University, Seoul, Korea

- ABEEK-accredited engineering program

PUBLICATIONS

PUBLISHED

[C1] FoodVitals: A Diagnostic Benchmark for LLMs on Nutritional Literacy, Nutrient–Health Relations, and Food Safety
CIKM 2026, Resource Track *Co-second author*
Benchmark of 1,158 food-knowledge items verified against USDA, EFSA, and FSIS sources, asking the same knowledge in four formats (MCQ, Yes/No, short answer, ranking). Evaluated six LLMs; on nutrient-content ranking, every model stays below 29% accuracy.

[C2] KARMA: Knowledge Graph-Based Automated Reasoning Materialization and Alignment
Findings of EMNLP 2026 *Second author*
Proposes contrastive training data synthesis based on knowledge graph paths and slot-level preference alignment (SPA).

UNDER REVIEW

[U1] Learning the Rate but Not the Days: Diagnosing and Repairing Time Series Foundation Models on Zero-Inflated E-Commerce Demand
Knowledge-Based Systems (Elsevier) ***First author***
Diagnostic framework that attributes forecast error on zero-inflated demand (37–56% zero days) to four measurable causes. No evaluated zero-shot TSFM (Chronos-2, Chronos-Bolt, Moirai-2, TimesFM-2.5) beats the classical SBA baseline on category-average MSE. Repairing only the fine-tuning objective of a retrieval-augmented TSFM stack closes its 3–20% deficit; the proposed SBA-ZI estimator attains lower pooled CRPS than every evaluated zero-shot TSFM. Released a new 31,318-series item-level panel.

[U2] KEEPER: Learning Control-Sufficient Textual Memory for Long-Horizon LLM Agents *Second author*
Defines self-consistency bias, where existing errors persist as an agent recursively updates its textual memory, and compares four auxiliary training objectives combined with offline RL on ScienceWorld and ALFWorld (Llama-3.2-3B).

KOREAN CONFERENCES

[D1] Mitigating Distribution Mismatch in Zero-Inflated Time Series Foundation Models: Mixture of Experts-Based Domain Adaptation and Log Transformation Strategies
KCC 2026 (Korea Computer Congress) ***First author***
MoE-module fine-tuning with log transformation, improving MSE by 7.5% on average.

[D2] An Efficient Travel Itinerary Generation Agent Based on LLMs with Plugin Tools and ReAct Orchestration
KICS Winter Conference 2026 *Co-author*
A ReAct-based travel-itinerary generation agent.

IN PREPARATION

[P1] Substitution-Aware Food Knowledge Graph and MoE-Based On-Device Food Recommendation Agent (working title)
In preparation ***Co-first author***
Generating the data that adds dietary restriction, substitute ingredient, and health score relations to a recipe knowledge graph (ongoing).

RESEARCH PROJECTS

Project Researcher, KIRD 2026 Global R&D Challenge (AdaptEat) Jun 2026 – Oct 2026
Korea–Uzbekistan food-KG-based substitute recommendation and explanations, with New Uzbekistan University

- Matched 1,434 ingredients with 572 FlavorDB flavour descriptors, generating 162,729 flavour edges (56% of all edges) in a Korea–Uzbekistan food knowledge graph.
- Implemented PathSim graph recommendation and its reciprocal rank fusion, improving Top-1 same-category results.
- Built the graph-grounded explanation verification module and the recommendation reasoning view in HTML/CSS.

Project Researcher, Yanolja Joint Research Sep 2025 – Aug 2026
Tourism data analysis and technology development: LLM travel-itinerary agent

- Fine-tuned KONI-Llama3.1-8B on Reddit travel data with QLoRA and RAFT and applied it as the planner of a ReAct travel-itinerary agent; traced budget and currency hallucinations to domain mismatch and overfitting in the training data.
- Ported the agent to a Seoul dataset (district-level location column, citySet).

HONORS AND LEADERSHIP

Lab Master, FNAI Lab, Sejong University Jun 2026 – present
Excellence Award, Multicultural Research Team Best Practice Contest (KIRD) Dec 2025

SKILLS

Programming	Python, PyTorch (primary); SQL, C++, HTML/CSS
Frameworks	Hugging Face Transformers, Git, LaTeX
Research	Knowledge graph data generation, LLM evaluation benchmark design, LLM agents (ReAct, tool use), time-series foundation models (Chronos, Moirai, TimesFM), MoE fine-tuning, knowledge-graph-based recommendation
Languages	Korean (native), English (professional working proficiency)